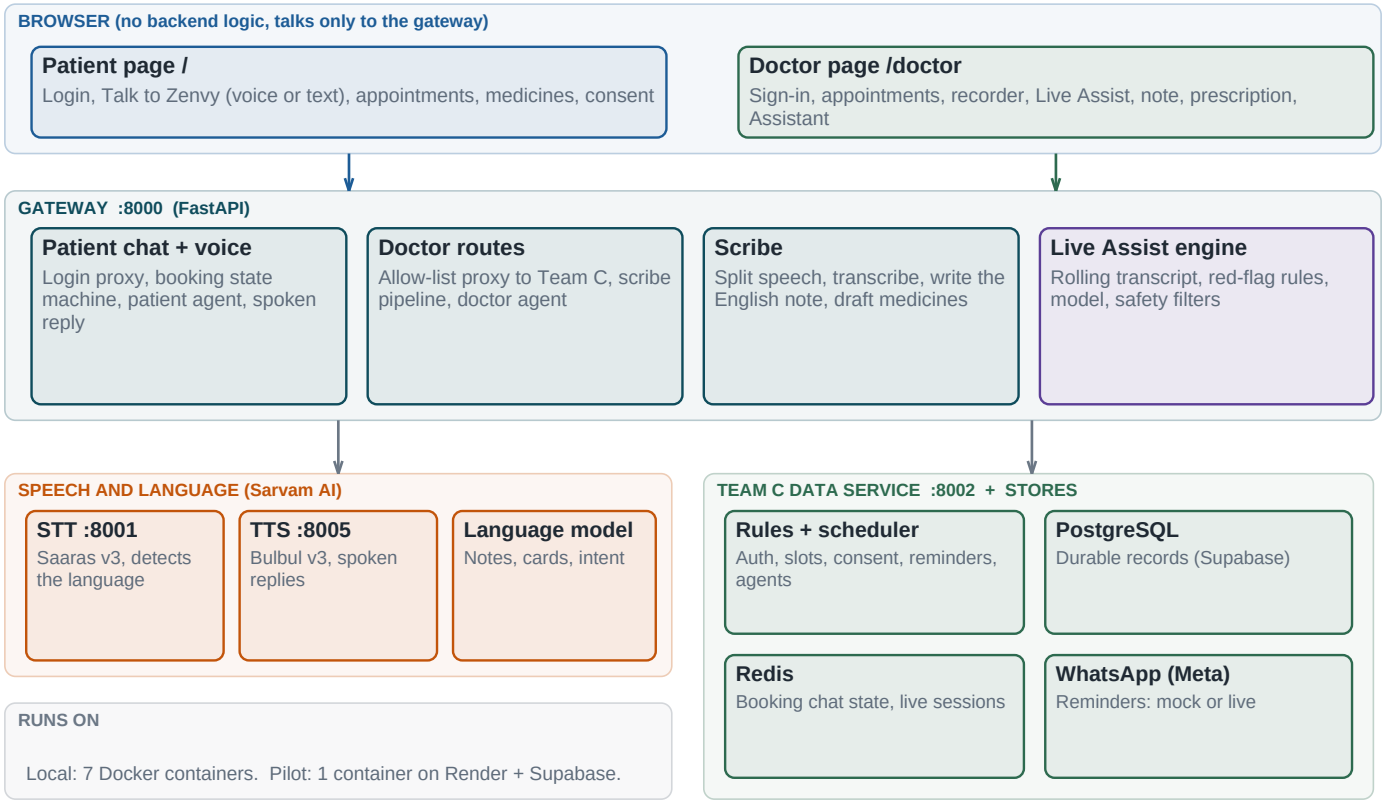
Zenvy Hospital Pilot: how it works

End-to-end workflow for the team: the patient page, the doctor page, Live Assist, and where the data goes. Describes what is built on branch **team-a/rishabh** (commit 4c6c11f).

1. The big picture

Two web pages share one gateway. The gateway does the AI work (speech, language models) and forwards everything else to **Team C's data service**, which owns the database and enforces the rules. AI only *drafts and suggests*; a person approves.



2. The rules the design follows

- **AI drafts, people decide.** Notes start as drafts; only the doctor approves, and an approved version is locked. Prescriptions are signed by the doctor; the agents never approve or sign.
- **The database enforces, not the page.** Slot locking, recording consent, and "only the treating doctor can open this consultation" are checked in Team C, so a modified browser cannot skip them.
- **The doctor page reaches only an allow-list** of Team C routes and carries the doctor's own token; Team C decides what that doctor may see.
- **Plain code does the work, models only understand and draft.** Booking, reminders and doses are scheduled by code, never by a model.

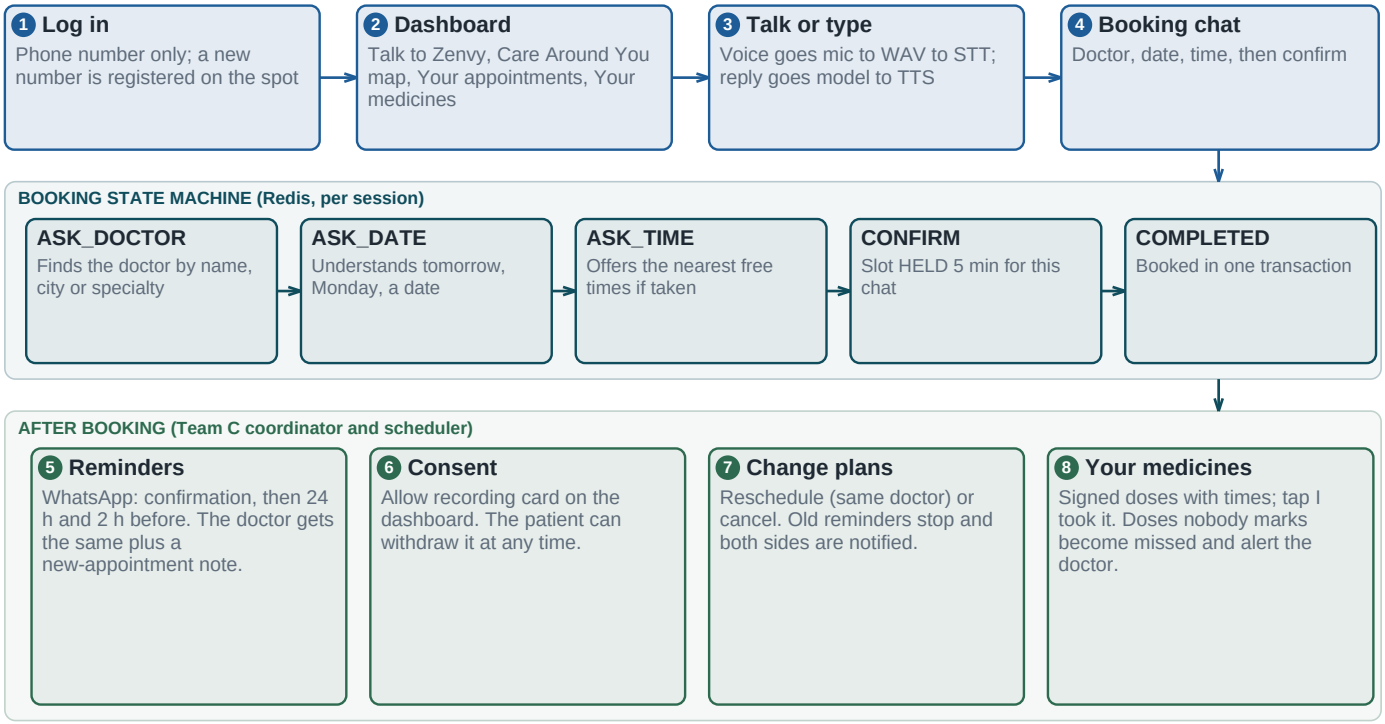
3. Components and settings

Component	Where	What it does
Gateway	`services/gateway/`	Serves both pages; patient chat and voice; doctor routes; calls STT, TTS and the language model.
Orchestrator	`services/orchestrator/`	Booking, cancel and reschedule state machine (state kept in Redis for 1 h).
Agents	`services/agents/`	Patient agent (medicine questions), doctor agent (Assistant panel), Live Assist engine.
Scribe	`services/scribe/`	Recording to transcript to English note; live sessions; medicine extraction.
Team C	`DB-Team-C/`	FastAPI + SQLAlchemy; 14 migrations; scheduler (reminders, agent events, daily purge).

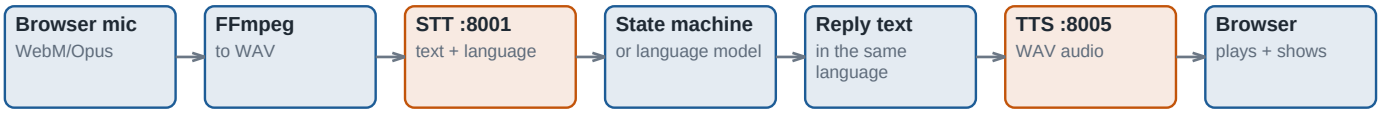
Setting	Default	Meaning
`REMINDER_MODE`	mock	mock records the message only; live sends WhatsApp templates
`SLOT_HOLD_SECONDS`	300	How long a patient's chosen slot is reserved
`AUDIO_RETENTION_DAYS / TRANSCRIPT_RETENTION_DAYS`	30 / 90	When the daily purge removes audio and transcripts
`AUDIO_ENCRYPTION_KEY`	required	32 random bytes; without it recording upload is refused

4. Patient side: from login to medicines

The patient uses one page at `/`. They can talk (Kannada, Hindi or English) or type; the reply comes back as text and spoken audio.



Voice path (same pipeline for every spoken request)

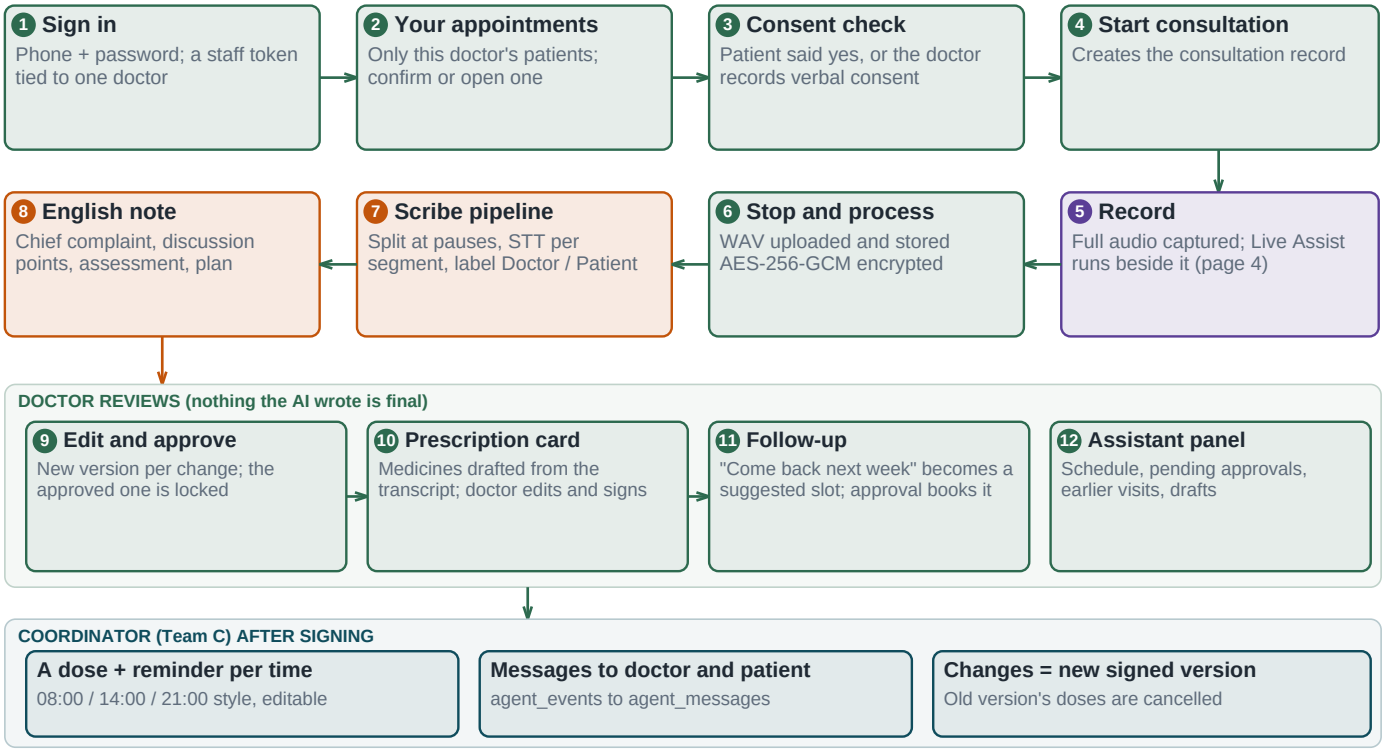


Step by step

#	Patient does	What happens behind the scenes
1	Enters a phone number.	`POST /channels/web/login` goes to Team C. Unknown numbers are registered as patients. No password or OTP yet.
2	Lands on the dashboard.	Loads their appointments and signed medicines from Team C; shows nearby hospitals and doctors by GPS or a demo city.
3	Speaks or types a request.	STT returns text and language; the state machine or model answers; TTS speaks it. Every turn is stored in Redis, then Postgres.
4	Chooses doctor, date, time.	Dates such as "next Monday" are resolved in IST. The doctor is looked up in Team C and free `doctor_slots` are read (made 14 days ahead from each weekly schedule).
4	Says yes at CONFIRM.	**Two atomic steps:** `POST /slots/{id}/hold` reserves the slot for this chat (5 min, others get 409); `POST /appointments` turns the hold into a booking. A unique index on `ai_appointments(slot_id)` is the last safety net.
5	Nothing.	Reminders are rows in `reminders`, sent by the scheduler every 30 s. One atomic UPDATE claims each, so two schedulers never double-send; a failure retries once after 5 min.
6	Taps Allow recording.	Stored as a versioned consent record (wording v1 in English, Hindi, Kannada). Recording cannot start without a current yes; a patient's refusal cannot be overruled by the doctor.
7	Reschedules or cancels.	Reschedule claims the new slot, creates the new appointment and cancels the old one in **one transaction**. Started appointments, and ones with a consultation, cannot be changed.
8	Asks "what medicines do I take?" or taps a dose.	The patient agent reads only the doctor-signed prescription. Unmarked doses become *missed* after 3 h; two misses in a day alert the doctor once.

5. Doctor side: from sign-in to a signed prescription

The doctor uses `/doctor`. Every action on the page goes through the gateway's allow-list; patients appear masked (`Patient ••••1234`), never with a full number.



What each stage guarantees

Stage	Behind the scenes	If something fails
Upload	`POST /doctor/api/consultations/{id}/audio`. The server re-checks consent, then encrypts with `AUDIO_ENCRYPTION_KEY`; refuses (503) when no key is set.	Consent not granted: rejected, nothing stored.
Transcribe	Audio is split on speech pauses; segments go to STT three at a time; turns are labelled Doctor or Patient from pause length.	*no_speech*, *speech_to_text_failed*, *audio_unreadable*, shown to the doctor.
Write note	The language model writes the note in English whatever language was spoken (translated if needed); long visits are split and merged.	*summary_failed*, *summary_not_english*. The doctor can Regenerate.
Medicines	A second pass drafts a medicine list. A medicine whose name is not in the transcript or plan is thrown away.	No draft: the doctor fills the card by hand.
Approve	Notes are versioned; approving locks that version. The audit log keeps what the AI drafted next to what was signed.	Edits after approval become a new version.
Follow-up	Only English time phrases ("in 2 weeks", "next week") are recognised; booked at the same time of day or the nearest free slot, with normal slot locking.	The doctor can book or change it manually.

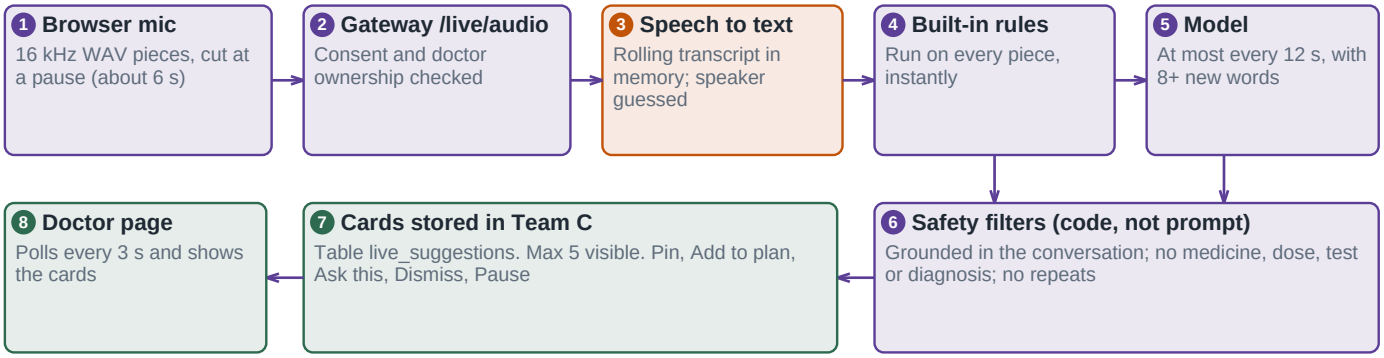
Also on the doctor page: the consent state of each appointment; **Delete recording and transcript** (notes are kept); **Regenerate note**; an appointment history that lists what happened and every message with its status; and the **Assistant panel** (doctor agent), which answers "what is my schedule?", "what is waiting for approval?" and "summarise this patient's earlier visits with me", and drafts a prescription or follow-up date. It reads and drafts; it never approves, signs or gives medicine advice.

One visit, end to end (example timeline)

When	What happens	Who or what does it
Before the visit	Slot held 5 min, then booked; WhatsApp confirmation; reminders at 24 h and 2 h; the patient taps Allow recording.	Patient chat, Team C, scheduler
Visit	Doctor starts the consultation and records; Live Assist cards appear as the talk goes on.	Doctor page, gateway, Live Assist
Minutes later	Recording stored encrypted, transcribed, written up as a draft note plus a draft medicine list.	Scribe pipeline
Same day	Doctor approves the note and signs the prescription; doses and reminders are scheduled; follow-up booked on approval.	Doctor, coordinator
Days after	The patient taps I took it; unmarked doses become missed and alert the doctor. Audio is purged at 30 days, transcript at 90.	Patient page, scheduler, daily purge

6. Live Assist: suggestions while the conversation is going on

While the doctor records, the same microphone also feeds **Live Assist**. It shows short English text cards to the **doctor only**, until the recording stops. Nothing is written into the note, prescription or patient record.



What the doctor sees (examples from a real test run)

RED FLAG
Fainting or loss of consciousness was mentioned - consider urgent evaluation. rule, within seconds

ASK NEXT
Ask about the exact duration of the fainting episode. because: "I fainted for a few seconds"

KEEP IN MIND
Keep in mind: stress from a recent job change may be a contributing factor. because: "I changed jobs four months ago"

Card kinds: ASK NEXT, RED FLAG, HISTORY (this doctor's earlier visits), WRAP-UP, KEEP IN MIND. A card appears about 10 to 20 s after the sentence; red-flag rules (chest pain, difficulty breathing, fainting, seizure, heavy bleeding, self-harm, stroke signs, worst headache, face or throat swelling) fire within seconds.

Safety rules in code: a card must quote words that were really said; cards naming a medicine, dose, test, disease or diagnosis are dropped; a model "red flag" without urgent wording is downgraded; the rules read statements, not the doctor's questions ("Any chest pain?"), and ignore denials ("no chest pain"). If speech or the model fails the panel says "Live Assist paused" and the recording and note are unaffected.

7. Where the data lives and for how long

Data	Where	Kept	Removed by
Audio recording	Encrypted file (AES-256-GCM)	30 days	Delete recording, consent withdrawal, daily purge
Transcript + Live Assist cards	consultation_turns, live_suggestions	90 days	Same three triggers
Notes and prescriptions	consultation_notes, prescriptions	7 years (to confirm)	Never by the system
Consent, audit trail, agent log	consultation_consents, audit_log, agent_*	7 years	Never by the system
Live audio pieces and live transcript	Gateway memory only	The visit	Discarded, never saved

Who sees what: only the treating doctor opens a consultation (other doctors get 403); patients never read clinical data; no phone numbers in the audit log or the Live Assist table.

8. Built and tested, and what is not done yet

- **Built:** slot locking, three-agent system, consultation recording and scribe, follow-up and reminders, medication flow, Live Assist.
- **Tests:** 694 passed (root); Team C 338 on SQLite and 370 on PostgreSQL, including a 20-writer slot race.
- **Production now:** Supabase has migrations 001 to 013 and Render runs the pilot; Live Assist is local only for now.
- **Live Assist languages:** English only today. Kannada and Hindi are planned (translate speech to English first).
- **Login:** patient login is phone-number only; OTP is not built.
- **Before real patients:** Sarvam data-processing agreement, key rotation, native review of Hindi and Kannada consent text, drug-allergy checking.
- **Deploy:** migration 014 on Supabase; the Render free plan is too small for Live Assist.

9. Where to look and how to try it

Want to see	Look here
Pages	`http://localhost:8000/` (patient) and `/doctor`. Doctor sign-ins are seeded; passwords come from `SEED_DOCTOR_PASSWORD`.
Live Assist	`LIVE_ASSIST.md`, `services/agents/live_assist.py` (rules, filters), `services/scribe/live.py` (sessions). Data rules: `DB-Team-C/README.md`, `RETENTION_POLICY.md`.
Test audio	`scripts/live_samples/`: English and Kannada doctor-patient recordings (normal visit, red flag, no false alarm, hair fall, broken leg).
Tests	`python -m pytest` in the repo root and in `DB-Team-C`; Postgres tests need `TEST_POSTGRES_URL` pointing at a throwaway database.